```
ONBOOT=yes
NETWORK=192.168.140.0
NETMASK=255.255.255.0
IPADDR=192.168.140.251
BROADCAST=192.168.140.255
GATEWAY=192.168.140.1
USERCTL=no
```

Edit the configuration file for each network interface that you want to add in to the *bond0* group, similar to the following example (change the interface names to ones appropriate for your system). I will be adding *eth2* and *eth3* to *bond0*:

```
[root@bastion network-scripts]# cat ifcfg-eth2
DEVICE=eth2
ONBOOT=yes
TYPE=Ethernet
MASTER=bond0
SLAVE=yes
```

The *ifcfg-eth3* file's contents will be the same, except that the DEVICE name changes to *eth3*.

Now restart your network service, or just bring up the bonding device, if your administration tools allow it. Run *tail -f /var/log/*messages in a separate terminal window, to watch for bonding driver error messages.

The *messages* output below indicates that *bond0* has been successfully set up, with the two interfaces in slave mode:

```
[root@bastion ~]# tail -f /var/log/messages
Apr 14 10:59:35 bastion NetworkManager: <info>  Activation (eth2) Stage 5
of 5 (IP Configure Commit) started...
Apr 14 10:59:36 bastion NetworkManager: <info>  (eth2): device state
change: 7 -> 8 (reason 0)
Apr 14 10:59:36 bastion NetworkManager: <info>  Policy set 'System bond0'
(eth2) as default for routing and DNS.
Apr 14 10:59:36 bastion NetworkManager: <info>  Activation (eth2)
successful, device activated.
Apr 14 10:59:36 bastion NetworkManager: <info>  Activation (eth2) Stage 5
of 5 (IP Configure Commit) complete.
Apr 14 10:59:36 bastion NetworkManager: <info>  Activation (eth3) Stage 5
of 5 (IP Configure Commit) started...
Apr 14 10:59:37 bastion NetworkManager: <info>  Policy set 'System bond0'
(eth2) as default for routing and DNS.
Apr 14 10:59:37 bastion NetworkManager: <info>  (eth3): device state
change: 7 -> 8 (reason 0)
Apr 14 10:59:37 bastion NetworkManager: <info>  Activation (eth3)
successful, device activated.
Apr 14 10:59:37 bastion NetworkManager: <info>  Activation (eth3) Stage 5
of 5 (IP Configure Commit) complete.
Apr 14 11:00:44 bastion kernel: bonding: bond0: doing slave updates when
interface is down.
Apr 14 11:00:44 bastion kernel: bonding: unable to remove non-existent
slave eth2 for bond bond0.
```

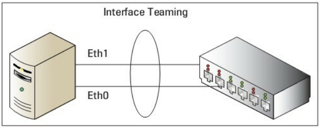

Figure 1: Bonded/teamed link from a server to an Ethernet switch

```
Apr 14 11:00:44 bastion NetworkManager: <info>  (eth2): carrier now OFF
(device state 8, deferring action for 4 seconds)
Apr 14 11:00:45 bastion kernel: bonding: bond0: doing slave updates when
interface is down.
Apr 14 11:00:45 bastion NetworkManager: <info>  (eth3): carrier now OFF
(device state 8, deferring action for 4 seconds)
Apr 14 11:00:46 bastion kernel: lo: Disabled Privacy Extensions
Apr 14 11:00:46 bastion kernel: ADDRCONF(NETDEV_UP): bond0: link is not ready
Apr 14 11:00:46 bastion kernel: bonding: bond0: Bastion slave eth2.
Apr 14 11:00:46 bastion kernel: eth2: link up
Apr 14 11:00:47 bastion NetworkManager: <info>  (eth2): carrier now ON
(device state 8)
Apr 14 11:00:48 bastion kernel: bonding: bond0: enslaving eth2 as an active
interface with an up link.
Apr 14 11:00:48 bastion kernel: ADDRCONF(NETDEV_CHANGE): bond0: link
becomes ready
Apr 14 11:00:48 bastion kernel: bonding: bond0: Bastion slave eth3.
Apr 14 11:00:48 bastion kernel: eth3: link up
Apr 14 11:00:48 bastion NetworkManager: <info>  (eth3): carrier now ON
(device state 8)
```

In the latest Linux distributions that support init scripts, it is possible to also specify options for the bonding driver in the *ifcfg-bond0* file, in a single line—for example:

```
BONDING_OPTS="mode=1 arp_interval=60 miimon=0"
```

If the system doesn't support *BONDING_OPTS*, you must modify */etc/modules.conf* or */etc/modprobe.conf* to add the following lines:

```
alias bond0 bonding
options bond0 mode=0 miimon=100
```

The above creates the *bond0* interface, and sets the options provided. It will check the MII state of the card every 100 milliseconds, for any state change.

Possible bonding module options

- ***mode:*** values could be 0 for a round-robin policy (the default), or 1, which can act like an active back-up. The round-robin policy transmits packets in sequential order from the first available slave through the last. This mode provides load balancing and fault tolerance. The active back-up or